

CCP6224 Object Oriented Analysis and Design

Mini Lab 3b – JavaFX Hello World

Lab outcomes

By the end of today's lab, you should be able to

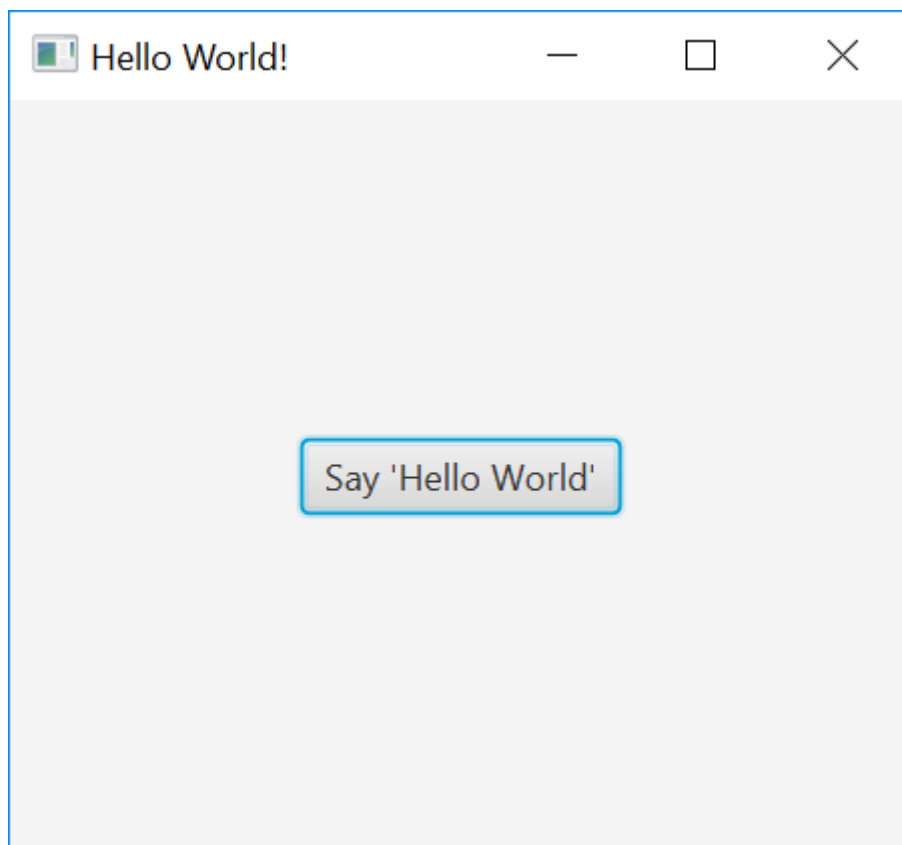
- Create your first JavaFX program in BlueJ 4.0 or above

Note

Starting from BlueJ version 4.0, JavaFX is automatically supported. You just have to create your program in BlueJ the normal way, and if it is a JavaFX program, it will have a “run JavaFX Application” menu item on every class that subclasses `javafx.application.Application`.

Exercise

This exercise will create GUI like this:



When the button is pressed, it just prints “Hello World!” to the terminal window.

JavaFX applications do not need `public static void main(String[] args)` – instead, it uses `public void start(Stage primaryStage)`. It is *allowed* to have a `main`, but it is not required to.

However, it is useful to include the `main()` method so you can run JAR files that were created without the JavaFX Launcher, such as when using an IDE in which the JavaFX tools are not fully integrated. Also, Swing applications that embed JavaFX code require the `main()` method.

https://docs.oracle.com/javafx/2/get_started/hello_world.htm

The main GUI class should subclass `javafx.application.Application`. Here is the sample `HelloWorld.java` code:

```
import javafx.application.Application;
import javafx.event.ActionEvent;
import javafx.event.EventHandler;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.layout.StackPane;
import javafx.stage.Stage;

public class HelloWorld extends Application {
    public static void main(String[] args) {
        launch(args);
    }

    @Override
    public void start(Stage primaryStage) {
        primaryStage.setTitle("Hello World!");
        Button btn = new Button();
        btn.setText("Say 'Hello World'");
        btn.setOnAction(new ButtonPressed());

        StackPane root = new StackPane();
        root.getChildren().add(btn);
        primaryStage.setScene(new Scene(root, 300, 250));
        primaryStage.show();
    }

    private class ButtonPressed implements EventHandler<ActionEvent>
    {
        @Override
        public void handle(ActionEvent event) {
            System.out.println("Hello World!");
        }
    }
}
```

Enter the above code into a new class in BlueJ, compile it, and select the “run JavaFX Application” menu item on that class.